

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Title – CareerSathi AI

AI-Powered Career Guidance and Skill Development Assistant

Domain of Project – Education

Student Name	Recent Passport Photo	Email ID	Mobile number [WhatsApp]
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Problem Statement - AI Career Counselor Rural Youth

The Challenge:

Many rural students face difficulties in choosing the right career path due to **limited access to professional career guidance**, **lack of awareness about emerging opportunities**, and scattered information about skills, courses, and job requirements.

As a result, **students often struggle to make informed decisions about their education**, skill development, and future careers.

The Objective:

Our Solution – CareerSathi AI

CareerSathi AI is an AI-powered career guidance assistant that provides personalized career recommendations, skill-development roadmaps, learning resources, and placement preparation guidance.

The platform helps students:

- ▶ Explore suitable career paths
- ▶ Identify required skills and learning resources
- ▶ Build personalized career roadmaps
- ▶ Receive placement and interview guidance
- ▶ Access reliable career information anytime

Impact

CareerSathi **AI empowers rural youth with accessible**, personalized, and real-time career guidance, helping them transform career aspirations into actionable plans.

Technology Used

Tech Stack

Core Technologies

IBM watsonx Orchestrate – Agent creation, management, and deployment

IBM watsonx.ai – AI platform for model access and inference

IBM Granite Models – Natural language understanding and career guidance generation

IBM Knowledge Base – Stores and retrieves career knowledge for accurate responses

Supporting Technologies

Vercel – Web hosting and deployment

HTML – Frontend structure

JavaScript – Agent embedding and interaction

Knowledge Base / RAG – Career information retrieval and recommendations

IBM watsonx Orchestrate + IBM watsonx.ai + IBM Granite Models + Vercel

Proposed Solution -

CareerSathi AI is an **AI-powered Career Counseling Agent** built to empower rural and underserved students with personalized career guidance. It acts as a **virtual career mentor** that understands a student's interests, strengths, academic background, and career goals to recommend suitable career paths, skills to learn, and opportunities to explore.

Built using **IBM watsonx Orchestrate, IBM watsonx.ai, Granite Models, and LangFlow**, the solution combines intelligent conversation, knowledge retrieval, and multi-agent orchestration to deliver accurate, personalized, and actionable career guidance.

Key Agents & Capabilities



Career Discovery Agent
Identifies suitable career paths based on student interests, strengths, personality, and aspirations.



Skill Gap Agent
Analyzes current skills against industry demands and identifies skill gaps with improvement tips.



Learning Roadmap Agent
Creates personalized learning roadmaps with courses, resources, certifications, and projects.



Placement Support
Provides interview preparation, resume building tips, and job search strategies.

Key Features

- Natural language conversation in English & regional languages
- Personalized career recommendations
- Skill assessment & gap analysis
- Step-by-step learning roadmaps
- Course & certification recommendations
- Resume & interview preparation support
- Scholarship & higher education guidance
- Internship & job opportunity suggestions
- Real-time doubt resolution
- Progress tracking & goal management

Usage

Students interact via chat interface
→ share their skills interests, skills, education background → receive personalized career options
→ follow learning roadmap → build skills → get placement & higher education guidance.

CareerSathi AI

Agent chat / Manage agents / CareerSathi AI

Test agent

live

Deploy



CareerSathi AI



Model : GPT-OSS 120B — Ope...

Build with AI

Talk to agent



Running in debug mode

1:04 PM



From ambition to action—let's build your career roadmap

Accuracy of generated answers may vary. Please double-check responses.



Discover a career you may ha...



What opportunities am I missing right...



Turn my interests into a rea...



Type something...



Profile

Knowledge

Toolset

Behavior

Channels Preview



Notifications: 1 Dismissed | 1 total



Legal Notice +

View all



Profile



Describe what your agent does so people and other agents know when to use it.

Description*

CareerSathi AI is an intelligent career guidance assistant designed to help rural youth discover suitable career paths, identify skill gaps, explore learning opportunities, and access relevant government support programs.

The agent interacts with users in simple, easy-to-understand language and provides personalized career recommendations based on education level, interests, goals, location, and available resources. It can generate career roadmaps, suggest job-oriented skills, recommend free learning platforms, and

Describe your agent, including its purpose, tools, and agents, to help other agents and users know when to use it.

What makes a good description? ?

User Query Processing and Response Testing

IBM watsonx Orchestrate

EnglishContact SalesBuy Now25 trial days left

Agent chat / Manage agents / CareerSathi AI

CareerSathi AI

AI Model : GPT-OSS 120B — Ope...

Build with AITalk to agent

ProfileKnowledgeToolsetBehaviorChannels Preview

Notifications: 1 Dismissed | 1 total

Legal Notice +

View all

Profile

Describe what your agent does so people and other agents know when to use it.

Description*
roadmaps, suggest job-oriented skills, recommend free learning platforms, and guide users toward scholarships, training programs, and skill development initiatives.

CareerSathi AI aims to bridge the career awareness gap by making professional guidance accessible, personalized, and available anytime, helping students and job seekers make informed decisions about their future.

Describe your agent, including its purpose, tools, and agents, to help other agents and users know when to use it.

What makes a good description?

Running in debug mode

Today

You 12:51 PM

I am a 2nd year B.Tech CSE student interested in Artificial Intelligence and Machine Learning. Based on my profile, suggest suitable career paths, required skills, salary ranges, and a roadmap for the next 2 years. Analyze my current skills in Python, HTML, CSS, SQL, and basic Machine Learning and identify the skill gaps I need to fill to become an AI Software Engineer. Create a month-by-month learning roadmap covering DSA, Machine Learning, Deep Learning, Generative AI, AWS, MLOps, and system design. Recommend internships that match my profile and explain how they will help my career. Review my resume and identify weak sections, missing skills, and improvements needed for AI-related internships and placements. Suggest 5 high-impact AI projects with technologies, difficulty levels, implementation steps, and expected learning outcomes. Generate 20 technical interview questions with answers for AI Engineer internships. Recommend the best free and paid resources for learning AI, ML, Deep Learning, AWS, and Generative AI. Compare AI Engineer, Machine Learning Engineer, Data Scientist, and Data Analyst roles in terms of skills, salary, demand, and future opportunities. Create a personalized weekly study plan

Save as test

Type something...

Generated Responses

1 Suitable AI-Career Paths

Path	Typical Work	Why it fits you	Core Tech Stack
AI Software Engineer (Product-focused)	Design & implement AI features inside web/mobile products; write production-grade code, integrate ML models via APIs.	You already know Python, Git, basic AWS, and want to ship real-world AI.	Python, FastAPI/Flask, Docker, CI/CD (GitHub Actions), AWS (Sage-Maker, Lambda), React/Angular (optional UI)
Machine Learning Engineer	Build, train, optimize, and serve ML models; focus on pipelines, scaling, monitoring.	Your ML basics + desire for deployment align well.	PyTorch/TensorFlow, MLflow, Kubeflow, Docker, AWS/GCP, SQL/NoSQL
AI Backend Developer	Create backend services that expose AI capabilities (recommendation, vision, NLP) to front-ends.	Emphasis on backend + AI.	Node.js/Express or Python FastAPI, REST/GraphQL, Redis, PostgreSQL, AWS Lambda, Docker
MLOps Engineer	Automate the end-to-end ML lifecycle (CI/CD for models, monitoring, data versioning).	You like "deployment" more than pure research.	Terraform, Docker, Kubernetes, MLflow, DVC, AWS SageMaker, CI pipelines
AI-Product Engineer (Startup-focused)	Wear many hats: prototype, build, test, and ship AI features quickly.	Good for Hyderabad's growing startup ecosystem.	Same as AI Software Engineer + rapid prototyping tools (Streamlit, Gradio)

2 Skill-Gap Analysis

Current Skill	Required Level for Target Role	Gap	How to Close (2 hrs/week)
Python (basics)	Intermediate-advanced (clean code, OOP, async)	Moderate	Daily coding challenges on LeetCode (easy/medium).
Git / GitHub	Proficient (branching, PR workflow, CI)	Small	Build a personal repo with CI using GitHub Actions (auto-run tests).
AWS (S3, EC2 basics)	Production-grade services (Lambda, SageMaker, IAM, CloudWatch, CI/CD)	Large	Follow AWS Free Tier labs – 1 hr/week on “Deploy a Flask app with Elastic Beanstalk”.
SQL	Advanced queries, indexing, ORM (SQLAlchemy)	Small	Add DB layer to a mini-project (e.g., blog).
HTML/CSS	Basic – okay for prototypes; not core.	None	Keep for quick UI demos (Streamlit/Gradio).
Machine Learning (theory + scikit-learn)	Deep learning, model serving, performance tuning	Large	Complete a focused DL course (see resources).
Deep-Learning Frameworks	PyTorch or TensorFlow (model building, custom layers)	None (assumed missing)	2 weeks hands-on PyTorch tutorial (official docs).
Backend Development	FastAPI/Flask, REST, Docker, testing, logging	Moderate	Build a CRUD API with Docker and unit tests.
MLOps basics	Docker, Kubernetes (minikube), MLflow, DVC, CI/CD for models	Large	Follow “MLOps Zoomcamp” (free).
Soft Skills	Technical communication, product thinking, agile basics	Small	Contribute to open-source, write project READMEs, attend local meetups (e.g., Hyderabad AI Club).

3 Month-by-Month Roadmap (10 hrs/week)

Month	Goal	Weekly Breakdown (5 hrs Sat, 5 hrs Sun)
1	Foundations – Python & Git	Sat: 2 hrs – Python OOP + typing; 3 hrs – LeetCode easy. Sun: 2 hrs – Git workflow + GitHub Actions; 3 hrs – Build a simple CLI tool and push to GitHub.
2	Web Backend + Docker	Sat: 3 hrs – FastAPI basics; 2 hrs – Docker intro. Sun: 3 hrs – Build CRUD API (PostgreSQL) + Dockerize; 2 hrs – Write unit tests (pytest).
3	Deep Learning Basics	Sat: 2 hrs – Linear algebra refresher (Khan Academy). Sat: 3 hrs – PyTorch official tutorial (Tensors, Autograd). Sun: 5 hrs – Complete “Deep Learning Specialization – Course 1 (Neural Networks)”.
4	Computer Vision / NLP Mini-Projects	Sat: 5 hrs – Build an image classifier (CIFAR-10) → serve via FastAPI. Sun: 5 hrs – Build a sentiment-analysis API (HuggingFace Transformers).
5	MLOps Foundations	Sat: 3 hrs – Docker + Docker-Compose for ML services. Sat: 2 hrs – DVC basics (data versioning). Sun: 5 hrs – “MLOps Zoomcamp” – Modules 1-2 (MLflow, CI for models).
6	AWS Production Services	Sat: 3 hrs – AWS IAM & CLI setup. Sat: 2 hrs – Deploy FastAPI on Elastic Beanstalk. Sun: 5 hrs – Deploy a model on SageMaker (SageMaker-scikit-learn or SageMaker-PyTorch).
7	Scaling & Monitoring	Sat: 2 hrs – CloudWatch alarms; 3 hrs – Logging (structured JSON). Sun: 5 hrs – Set up Prometheus + Grafana on Minikube to monitor model latency.
8	Advanced DL + Model Optimization	Sat: 5 hrs – Transfer learning (ResNet, BERT). Sun: 5 hrs – Quantization & TensorRT basics (optional).
9	Capstone Project – End-to-End AI Product	Sat: 5 hrs – Define product (e.g., “AI-powered job-matcher”). Sun: 5 hrs – Build backend, model pipeline, CI/CD, deploy on AWS (Lambda + API GW).
10	Portfolio & Resume Polish	Sat: 3 hrs – Write project READMEs, record demo videos. Sat: 2 hrs – Update LinkedIn & GitHub. Sun: 5 hrs – Tailor resume (see section 5) and apply to internships.
11-12	Internship Hunt & Interview Prep	5 hrs each weekend: practice coding (LeetCode medium), mock system-design, AI-role questions (see section 7).

4 Internship Recommendations (India, Hyderabad-friendly)

Company / Platform	Role (Typical)	Why a Good Fit	How to Apply
Infosys – AI & Automation	AI/ML Engineer Intern	Large campus hiring, strong mentorship, exposure to AWS & MLOps.	Apply via campus placement portal; also check Infosys Careers > “Internships”.
Tech Mahindra – Digital Engineering	ML Engineer Intern	Works on telecom AI products, good for backend + scaling.	Look for “Tech Mahindra AI Internship 2025” on Naukri/LinkedIn.
Zensar Technologies	AI Software Engineer Intern	Hyderabad office, focuses on product AI (chatbots, recommendation).	Submit resume + GitHub portfolio link.
Startups – CredR, Darwinbox, InMobi, Razorpay	AI Backend / MLOps Intern	Fast-paced, chance to ship features quickly.	Directly email founders/HR with a 1-page project showcase.
Research Labs – IISc/IIT-Hyderabad (ML/AI labs)	AI Engineer Intern	Good for solid fundamentals, often remote-friendly.	Check lab websites for “Summer Internship 2025”.
Freelance / Gig Platforms (Upwork, Toptal)	Small AI-API projects	Build real client work, earn while learning.	Create a profile, list your capstone project as a sample.

Action: Prepare a one-page “Project Portfolio” (title, problem, tech stack, outcome, link to repo) and attach it to every application.

5 Resume Improvement Suggestions

Section	Current (assumed)	Recommended Format
Header	Name, phone, email	Add LinkedIn & GitHub URLs, location (Hyderabad).
Objective	“Seeking AI role”	“B.Tech CSE student with 1yr Python/ML experience, building production-grade AI services on AWS. Looking to contribute as an AI Software Engineer.”
Education	B.Tech CSE – 8.2 GPA	Keep, add expected graduation (May 2026).
Technical Skills	List basics	Group by Languages, Frameworks, Cloud / DevOps, Tools . Use proficiency levels (e.g., Python – Advanced, PyTorch – Intermediate).
Projects	Probably 2-3 school projects	Expand to 5 items: 1 Image-Classifer API (FastAPI, Docker, AWS Elastic Beanstalk) – GitHub link 2 Sentiment-Analysis micro-service (Transformers, Lambda) – Live demo 3 End-to-End Job-Matcher (ML pipeline, CI/CD) – Video 4 Open-source contribution (e.g., bug fix in scikit-learn) 5 Hackathon/college-team project
Experience	Part-time / campus work (if any)	Quantify impact: “Reduced inference latency by 30 % using model quantization.”
Certifications	None (assume)	Add free certifications (e.g., “AWS Cloud Practitioner – Free tier”, “DeepLearning.AI TensorFlow Developer – Coursera (audit)”).
Achievements	GPA only	Mention any hackathon rank,  “Novice” badge, campus tech-club leadership.

AI Software Engineer (Product-focused) – This role blends your current strengths (Python, Git, basic AWS) with the product-side work you love. You'll design and ship AI-powered features, write clean production code, and deploy models on cloud services. It gives you direct ownership of the end-to-end flow (data → model → API → user) while keeping the work engineering-heavy rather than research-heavy.

Why it fits you

- **Current skill match** – you already know Python, Git, and have a starter grasp of AWS.
- **Growth potential** – you'll quickly pick up FastAPI, Docker, CI/CD, and MLOps as you build real products.
- **Job market** – Hyderabad's startups and large tech firms are actively hiring AI Software Engineers for chat-bots, recommendation engines, and vision APIs.
- **Product impact** – you get to see a feature go live and affect users, which aligns with your goal of building real-world AI applications.

Best Portfolio Project

AI-Powered Image Search Engine – Build a web service where users upload an image and receive visually similar product recommendations.

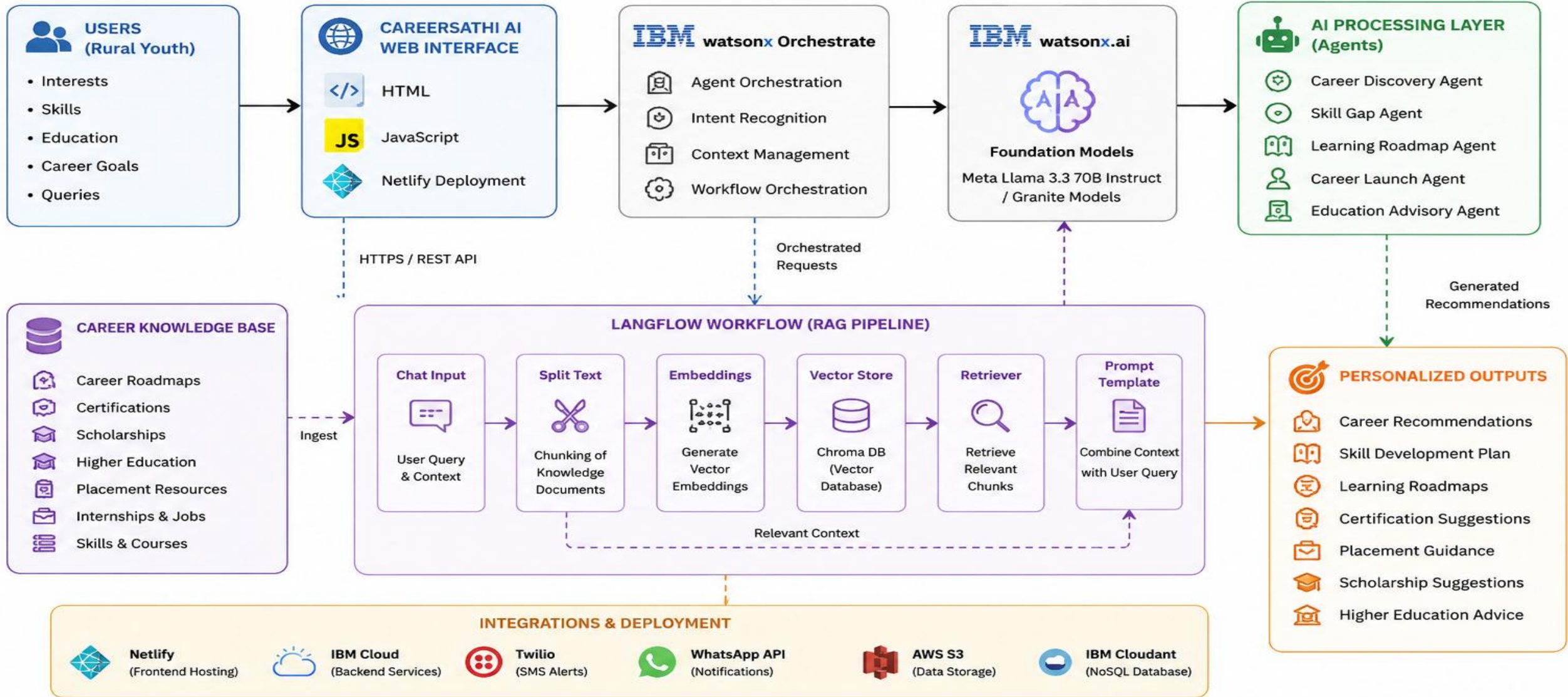
Key technologies you'll showcase

- **Model** – a CNN (e.g., ResNet-50) fine-tuned on a product-image dataset.
- **Backend** – FastAPI exposing `/search` endpoint.
- **Containerisation** – Dockerfile + Docker-Compose for reproducible deployment.
- **Cloud** – Store images in AWS S3; host the API on an Elastic Beanstalk or SageMaker endpoint; use AWS CloudWatch for logging and alerts.
- **Search** – Elasticsearch (or OpenSearch) with vector similarity for fast nearest-neighbor lookup.
- **Ops** – GitHub Actions CI pipeline that runs unit tests, builds the Docker image, and pushes it automatically on every push to `main`.



CareerSathi AI Architecture

AI-Powered Career Guidance Platform for Rural Youth



Role of Agentic AI in the solution

Role of Agentic AI in CareerSathi AI

Agentic AI enables CareerSathi AI to function as an intelligent, autonomous career guidance system rather than a simple chatbot. Using **IBM watsonx.ai**, **IBM watsonx Orchestrate**, and **Granite Models**, it coordinates multiple specialized agents to deliver personalized career recommendations.

Each agent performs a specific role—such as career discovery, skill gap analysis, learning roadmap generation, placement support, and higher education guidance—while working collaboratively. This **multi-agent approach** ensures accurate and context-aware career assistance.

Agentic AI understands a student's interests, educational background, skills, and career goals to provide tailored recommendations. It supports **real-time adaptation**, continuously refining guidance based on user inputs and changing career aspirations.

Additionally, agents communicate with each other, enabling better decision-making and end-to-end career planning. Overall, Agentic AI makes CareerSathi AI proactive, personalized, and goal-driven, acting like a **24/7 digital career mentor** that helps students make informed career decisions and achieve their professional goals.



Technology Used

- **IBM watsonx Orchestrate** – For orchestrating AI agents, managing conversations, and coordinating career guidance workflows.
- **IBM watsonx.ai** – For deploying foundation models and generating personalized career recommendations and learning roadmaps.
- **IBM Granite Models / Llama Models** – For natural language understanding, reasoning, career counseling, and personalized guidance.
- **Knowledge Base (KB)** – Stores career paths, certifications, scholarships, placement resources, and higher education information for accurate recommendations.
- **RAG (Retrieval-Augmented Generation)** – Retrieves relevant career information from the Knowledge Base before generating responses, improving accuracy and reliability.
- **Vector Database (Chroma DB)** – Stores and retrieves career knowledge efficiently for RAG-based recommendations.
- **HTML, CSS & JavaScript** – Used to build the CareerSathi AI web interface.
- **Vercel** – Used for deploying and hosting the CareerSathi AI website.

Novelty and Uniqueness

The proposed **CareerSathi AI** stands out by combining **Agentic AI, RAG, Knowledge-Based Guidance, and Personalized Career Planning** into a single intelligent career mentoring platform.

Multi-Agent Intelligence – Multiple specialized agents collaborate to provide career discovery, skill gap analysis, roadmap generation, placement support, and higher education guidance.

RAG-Based Accuracy – Retrieves career information from a curated Knowledge Base before generating responses, ensuring reliable and up-to-date recommendations.

Personalized Career Roadmaps – Generates customized learning paths based on a student's interests, skills, academic background, and career goals.

Rural Youth Empowerment – Bridges the career guidance gap for students in rural and underserved areas by providing 24/7 access to expert career mentorship.

Skill Gap Identification – Analyzes current skills and recommends targeted learning resources, certifications, and projects to improve employability.

Career-to-Action Guidance – Goes beyond suggestions by providing step-by-step roadmaps, practical learning plans, and actionable career milestones.

Higher Education & Scholarship Support – Recommends suitable degrees, certifications, scholarships, and higher education opportunities.

Placement Readiness Assistance – Provides interview preparation, resume guidance, and job-oriented learning recommendations.

Scalable & Enterprise-Ready – Built using **IBM watsonx Orchestrate, IBM watsonx.ai, Granite Models, LangFlow, and RAG architecture**, ensuring reliability, scalability, and AI governance.

Impact: CareerSathi AI acts as a **smart digital career mentor**, helping students make informed career decisions, build relevant skills, and confidently achieve their professional goals.

Git Hub Link

Github Link:

<https://github.com/kala-123-gif/IBM>

Careersathi_ai Link:

<https://careersathiai.vercel.app/>

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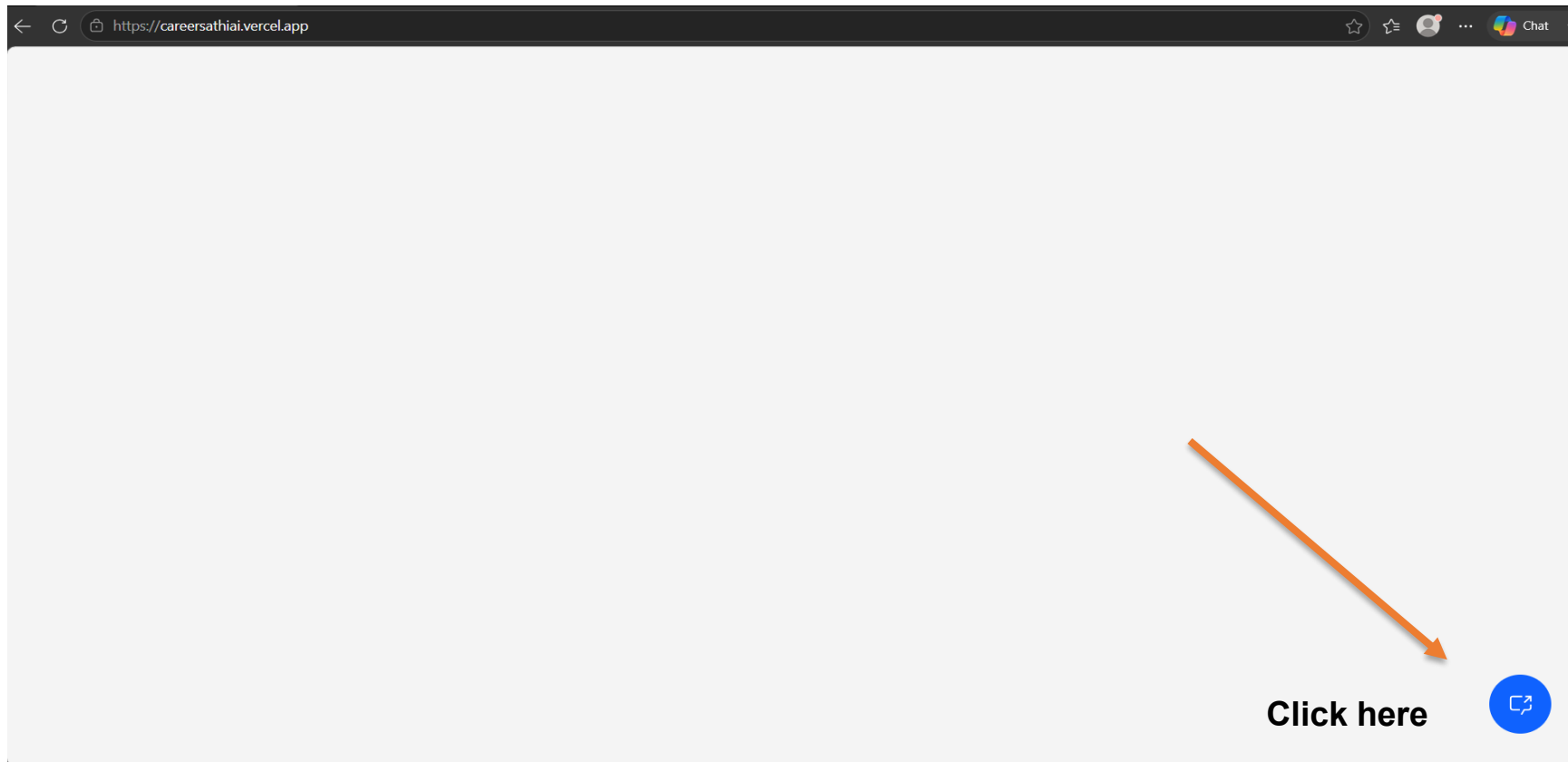
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How to Access CareerSathi AI

Step 1: Click the CareerSathi AI link.  [CareerSathi AI](https://careersathiai.vercel.app/) / <https://careersathiai.vercel.app/>

Step 2: The CareerSathi AI webpage will open.

Step 3: Click the chat toggle button located at the bottom-right corner of the page.

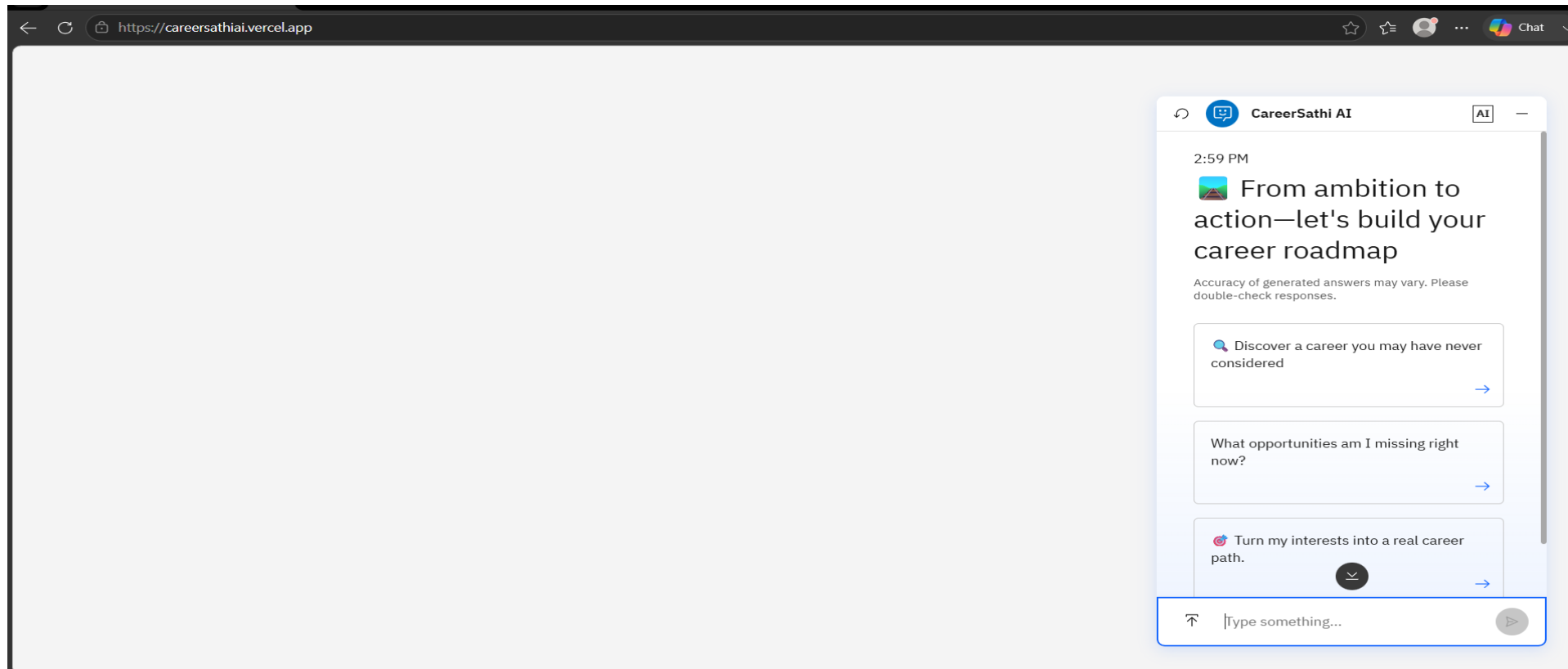


Using CareerSathi AI

Step 4: The CareerSathi AI chat window will open.

Step 5: Enter your career-related queries.

Step 6: Receive personalized career guidance, skill recommendations, learning roadmaps, and internship suggestions.



Token Usage

Resource usage

Usage summary

For this month in this project

0_{CUH}

7009 Tokens

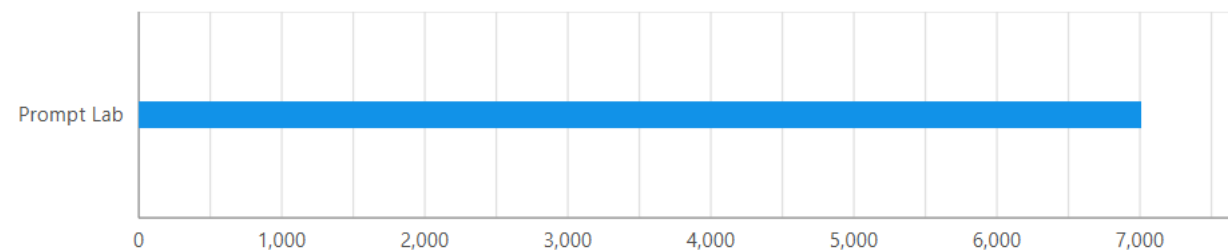
0 Hosting hours

0 Pages

[Manage service instances](#)

Usage by type

Tokens



Future Scope

1. AI-Powered Voice Assistant & Multilingual Support

The system can be enhanced with voice interaction and support for regional languages, enabling rural students to access career guidance in their preferred language. This will improve accessibility and inclusiveness.

2. Advanced Career Prediction & Job Market Insights

Future versions can integrate real-time industry trends, job market analytics, and skill demand forecasting to provide more accurate career recommendations and emerging career opportunities.

3. Personalized Learning & Certification Tracking

CareerSathi AI can track student progress, recommend personalized courses and certifications, and dynamically update learning roadmaps based on individual achievements.

4. Scholarship & Internship Recommendation Engine

The platform can automatically identify and recommend relevant scholarships, internships, competitions, and higher education opportunities based on student profiles.

5. Mobile App & Offline Accessibility

A dedicated mobile application with offline support can help students in rural and low-connectivity regions access career guidance anytime and anywhere.

6. Institution & Placement Cell Integration

CareerSathi AI can be integrated with colleges and placement cells to provide institution-specific career guidance, placement preparation, and student progress monitoring.

Thank You!

Thank you for your time and interest.